STPM Physics · Semester 2 · Chapter 18

ALTERNATING CURRENT CIRCUITS

18.1 AC Through a Pure Resistor

DEFINITIONS & LAWS

- An **alternating current** is a current that varies sinusoidally with time and reverses direction periodically, $I = I_0 \sin\omega t$.
- **Root-mean-square (r.m.s.) value** of an alternating current is the value of the steady direct current that dissipates heat in a given resistor at the same rate as the alternating current.
- In a pure resistor, the current is in phase with the voltage.
- The instantaneous power $P = I^2R$ varies with a frequency twice that of the current and is always non-negative; the mean power is half the peak power.
- Peak value: the maximum magnitude of the current or voltage during a cycle.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Current and voltage	$I = I_0 \sin\omega t$; $V = V_0 \sin\omega t$	I = current (A); V = voltage (V); I_0, V_0 = peak values; $\omega = 2\pi f$ (rad s^{-1}); f = frequency (Hz); t = time (s)
Period	$T = \frac{1}{f}$	T = period (s); f = frequency (Hz)
r.m.s. values	$I_{rms} = \frac{I_0}{\sqrt{2}}$; $V_{rms} = \frac{V_0}{\sqrt{2}}$	I_{rms} = r.m.s. current (A); V_{rms} = r.m.s. voltage (V); I_0, V_0 = peak values; sinusoidal waveform only
Peak current from r.m.s.	$I_0 = \sqrt{2}\, I_{rms} \approx 1.41\, I_{rms}$	I_0 = peak current (A); I_{rms} = r.m.s. current (A)
Mean power	$P = I_{rms}^2 R = \frac{V_{rms}^2}{R} = I_{rms} V_{rms}$ $= \frac{1}{2} I_0^2 R = \frac{1}{2} V_0 I_0$	P = mean power (W); R = resistance (Ω); I_{rms}, V_{rms} = r.m.s. current (A) and voltage (V); I_0, V_0 = peak values
Peak power	$P_{peak} = I_0^2 R = 2P_{mean}$	P_{peak} = peak power (W); P_{mean} = mean power (W); I_0 = peak current (A); R = resistance (Ω)
Energy dissipated	$E = P_{mean}\, t$	E = energy (J); P_{mean} = mean power (W); t = time (s)

HOW TO ANSWER (MPM STYLE)

- State the r.m.s. definition in terms of the heating effect of a steady d.c. in the same resistor.

- From $V = V_0 \sin(\omega t)$ identify V_0 as the coefficient of the sine and ω as the coefficient of t; then $f = \frac{\omega}{2\pi}$ and $V_{rms} = \frac{V_0}{\sqrt{2}}$.
- State $P = \frac{V_{rms}^2}{R}$ using r.m.s. values; if the peak value is given, convert first.
- For a sketch of V and I against t, show that the two curves are in phase, with labelled V_0, I_0 and period T.

SECTION A TRAPS

Trap: Using the peak value in $P = \frac{V^2}{R}$ gives the peak power, which is twice the mean power. Convert to r.m.s. first.

Trap: Reading ω (e.g. 314 or 377 rad s^{-1}) as the frequency in Hz. The frequency is $f = \frac{\omega}{2\pi}$ (50 Hz or 60 Hz).

Trap: Taking the mean voltage of a sinusoid as $\frac{V_0}{\sqrt{2}}$; the mean value over a full cycle is zero, and $\frac{V_0}{\sqrt{2}}$ is the r.m.s. value.

Trap: Not converting the time to seconds when calculating energy dissipated, for example 2.0 minutes = 120 s.

TRICKS

Trick: Mean power is always half of the peak power for a resistor: $P_{mean} = ½P_{peak}$. Use this to eliminate the option that equals the peak and the option that is a quarter of it.

18.2 AC Through a Pure Inductor

DEFINITIONS & LAWS

- **Inductive reactance** X_L is the opposition of an inductor to alternating current, given by the ratio of the r.m.s. (or peak) voltage across the inductor to the r.m.s. (or peak) current through it.
- In a pure inductor, the current lags the applied voltage by 90° ($\frac{\pi}{2}$ rad).
- The mean power dissipated in a pure inductor is zero over one complete cycle, because energy is stored in the magnetic field and returned to the source alternately.
- At very high frequency, the inductor opposes the current strongly; for direct current (f = 0), $X_L = 0$ and the inductor offers no opposition (apart from its resistance).

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Inductive reactance	$X_L = \omega L = 2\pi fL$	X_L = inductive reactance (Ω); ω = angular frequency (rad s^{-1}); f = frequency (Hz); L = inductance (H)
Current in an inductor	$I_{rms} = \frac{V_{rms}}{X_L}$; $I_0 = \frac{V_0}{X_L}$	I_{rms}, V_{rms} = r.m.s. current (A) and voltage (V); I_0, V_0 = peak values; X_L = inductive reactance (Ω)
Mean power in a pure inductor	$P = V_{rms} I_{rms} \cos\varphi = 0$	P = mean power (W); φ = phase angle = 90°; V_{rms}, I_{rms} = r.m.s. voltage (V) and current (A)

- X_L is directly proportional to f: a graph of X_L against f is a straight line through the origin with gradient $2\pi L$.

HOW TO ANSWER (MPM STYLE)

- For a calculation, state $X_L = 2\pi fL$, convert L to henry (mH → $\times 10^{-3}$), substitute and give the answer in Ω; then use $I = \frac{V}{X_L}$ with r.m.s. values.
- State the phase relation exactly: the current lags the voltage by 90° ($\frac{\pi}{2}$ rad), with the correct phrase for the inductor.
- To explain why the mean power is zero: energy is stored in the magnetic field in one quarter-cycle and returned to the source in the next, so the net energy transfer is zero.
- Sketch: sinusoidal V and I, same amplitude scale, with I displaced a quarter-period to the right of V.

SECTION A TRAPS

Trap: Stating that the current leads the voltage in an inductor; that is the capacitor. In an inductor, the voltage leads the current.

Trap: Using $X_L = \frac{1}{\omega L}$, inverted, or forgetting 2π (reading ω as f). Always check that X_L = 62.8 Ω for 0.20 H at 50 Hz.

Trap: Using the formula $P = I^2 X_L$ for the power, which suggests heating; the mean power is zero.

Trap: Calculating the current with the peak voltage and then quoting it as r.m.s. Match the peak with peak, r.m.s. with r.m.s.

TRICKS

> **Trick:** ELI the ICE man: in an inductor (L) the e.m.f. (E) leads the current (I); in a capacitor (C) the current leads the e.m.f.

> **Trick:** Doubling f doubles X_L and halves the current at a fixed supply voltage; use ratios to avoid recalculation.

18.3 AC Through a Pure Capacitor

DEFINITIONS & LAWS

- **Capacitive reactance** X_C is the opposition of a capacitor to alternating current, given by the ratio of the r.m.s. (or peak) voltage across the capacitor to the r.m.s. (or peak) current through it.
- In a pure capacitor, the current leads the applied voltage by 90° ($\frac{\pi}{2}$ rad).
- The mean power dissipated in a pure capacitor is zero over one complete cycle, because energy is stored in the electric field and returned to the source alternately.
- A capacitor blocks direct current ($X_C \to \infty$ as $f \to 0$) and passes high-frequency alternating current more easily ($X_C \to 0$ as $f \to \infty$).

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Capacitive reactance	$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$	X_C = capacitive reactance (Ω); ω = angular frequency (rad s^{-1}); f = frequency (Hz); C = capacitance (F)
Current in a capacitor (r.m.s.)	$I_{rms} = \frac{V_{rms}}{X_C}$	I_{rms} = r.m.s. current (A); V_{rms} = r.m.s. voltage (V); X_C = capacitive reactance (Ω)
Current in a capacitor (peak)	$I_0 = \frac{V_0}{X_C} = \omega C V_0$	I_0 = peak current (A); V_0 = peak voltage (V); X_C = capacitive reactance (Ω)
Reactance and frequency	$X_C \propto \frac{1}{f}$	X_C is inversely proportional to f; a graph of X_C against f is a decreasing curve (rectangular hyperbola) approaching zero at high frequency
Mean power in a capacitor	$P = V_{rms} I_{rms} \cos\varphi = 0$	P = mean power (W); φ = phase angle between V and I = 90° for a pure capacitor, so $\cos\varphi = 0$

HOW TO ANSWER (MPM STYLE)

- For a calculation, state $X_C = \frac{1}{2\pi f C}$, convert C to farad (μF → $\times 10^{-6}$), substitute and give the answer in Ω; then use $I = \frac{V}{X_C}$ with r.m.s. values.

- State the phase relation exactly: the current leads the voltage by 90° ($\frac{\pi}{2}$ rad).
- To explain the variation of current with frequency: as f increases, X_C decreases, so the current increases for a constant r.m.s. voltage.
- On a sketch of X_C against f, label the axes and show the curve falling steeply and approaching, but never touching, the f axis.

SECTION A TRAPS

Trap: Not converting μF to F; 10 μF = 10×10^{-6} F. An answer that is 10^6 times too large or too small indicates this slip.

Trap: Writing $X_C = 2\pi fC$ (inverting); X_C decreases with frequency, but X_L increases.

Trap: Stating that the current in the capacitor increases because the reactance increases; the reactance decreases as f increases.

Trap: Giving the capacitor current at $f \to 0$ as a maximum; $X_C \to \infty$, so the current is zero for steady d.c.

TRICKS

Trick: Frequency doubled: X_L doubles but X_C halves. Both effects are reciprocal; use ratios to check the numerical answer quickly.

18.4 R-C and R-L Series Circuits

DEFINITIONS & LAWS

- **Impedance** Z of an a.c. circuit is the ratio of the r.m.s. (or peak) voltage across the circuit to the r.m.s. (or peak) current in it.
- In a series R-L circuit, the voltage leads the current by a phase angle φ; in a series R-C circuit, the current leads the voltage by φ, where $0° < \varphi < 90°$.
- **Power factor** is the ratio of the mean power dissipated to the product of V_{rms} and I_{rms}, equal to $\cos\varphi = \frac{R}{Z}$.
- The mean power is dissipated only in the resistance; the inductor and capacitor dissipate no mean power.
- The voltages across R and L (or C) are 90° out of phase, so the supply voltage is their phasor (vector) sum and not their arithmetic sum.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
R-L series impedance	$Z = \sqrt{R^2 + X_L^2}$	Z = impedance (Ω); R = resistance (Ω); X_L = inductive reactance (Ω)
R-L series phase angle	$\tan\varphi = \frac{X_L}{R}$	φ = phase angle between voltage and current (rad or °)
Inductive reactance	$X_L = 2\pi fL$	X_L = inductive reactance (Ω); f = frequency (Hz); L = inductance (H)
R-C series impedance	$Z = \sqrt{R^2 + X_C^2}$	Z = impedance (Ω); R = resistance (Ω); X_C = capacitive reactance (Ω)
R-C series phase angle	$\tan\varphi = \frac{X_C}{R}$	φ = phase angle between voltage and current (rad or °)
Capacitive reactance	$X_C = \frac{1}{2\pi fC}$	X_C = capacitive reactance (Ω); f = frequency (Hz); C = capacitance (F)
Supply voltage and current	$V_{rms} = I_{rms} Z$	V_{rms} = r.m.s. supply voltage (V); I_{rms} = r.m.s. current (A); Z = impedance (Ω)
Voltage relation (phasor addition)	$V^2 = V_R^2 + V_L^2$ (or V_C^2); $V_R = IR$, $V_L = IX_L$, $V_C = IX_C$	V = supply voltage (V); V_R, V_L, V_C = voltages across R, L, C (V); I = current (A)
Power factor	$\cos\varphi = \frac{R}{Z}$	cosφ = power factor (no unit); R = resistance (Ω); Z = impedance (Ω)
Mean power	$P = V_{rms} I_{rms} \cos\varphi = I_{rms}^2 R$	P = mean power (W); V_{rms}, I_{rms} = r.m.s. voltage (V) and current (A); R = resistance (Ω)

HOW TO ANSWER (MPM STYLE)

- Draw the impedance (or phasor) triangle: R horizontal, X vertical, Z hypotenuse, and φ between R and Z; then use Pythagoras and trigonometry.
- Calculate X_L or X_C first, then $Z = \sqrt{R^2 + X^2}$, then $I_{rms} = \frac{V_{rms}}{Z}$ and finally the required voltage or power, giving each answer with a unit.
- State the phase relation clearly: 'current leads voltage by φ' for R-C, 'current lags voltage by φ' for R-L, with $\tan\varphi = \frac{X}{R}$.
- For the variation with frequency: R-L: as f increases, X_L and Z increase, so I decreases; R-C: as f increases, X_C decreases, so I increases.

SECTION A TRAPS

Trap: Adding arithmetically: Z = R + X (30 + 40 = 70 Ω). The correct value is Z = $\sqrt{30^2 + 40^2}$ = 50 Ω.

Trap: Using $\tan\varphi = \frac{R}{X}$ (inverted). For both circuits, $\tan\varphi = \frac{X}{R}$; check that φ = 90° when R = 0.

Trap: Taking the power factor as $\frac{Z}{R}$, which exceeds 1; the power factor is $\frac{R}{Z}$ and never exceeds 1.

Trap: Adding the component voltages to get the supply voltage; $V_R + V_L$ is greater than the supply, since $V = \sqrt{V_R^2 + V_L^2}$.

TRICKS

Trick: Recognise Pythagorean triples: (30, 40, 50), (60, 80, 100) and (6, 8, 10); Z, I and the voltage across each component then follow without a calculator.

Trick: Limiting case: when R increases with X fixed, φ → 0° and cosφ → 1 (more resistive); when R → 0, φ → 90° (purely reactive).